Utility Alerts — Stage 2 Sign-off Summary (Shadow Evidence Review)

Thesis link: §5.4 limitation "Official utility alerts may arrive later than primary-source announcements"; §5.5 future work #3 "real-time official alerts". **Date:** 2026-10-12 (FYP 2, Week 2) **Purpose:** the owner-facing evidence pack for the Stage 2 go/no-go decision (doc 04 §6). All numbers below are queried live from the production Supabase `utility_alert_shadow_log` / `utility_alerts` / `utility_alert_delivery` tables (deploy v116, shadow mode on).

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1. Evidence window

• Shadow tables active **2026-10-11 13:10 UTC** → **2026-10-12 10:47 UTC** (~21.6 h, 132 poll cycles). The earlier 10-06 shadow evaluation (doc 04 §4) covers detection correctness; this window covers the durable per-cycle evidence.

2. Acceptance criteria verdicts (doc 04 §5)

#	Criterion	Verdict	Evidence
1	Precision ≥ 90% (≥20 stored alerts)	[OK] **100% on sample (20/20)**	all stored rows are genuine Sarawak **water** disruption notices: 16 JBALB .gov.my (14 distinct locations statewide) + 4 official `swbnews` Telegram. See §4 for the review list. Caveat: power/South-North water remain news-mediated (doc 04 §2) — zero such alerts stored in this window, so precision *for those sources is untested*.
2	Zero pushes	[OK]	`utility_alert_delivery` row count = **0** while shadow mode on.
3	Quiet hours hold (00:00–06:00 Kuching)	■ **mechanism present, untested by data**	no alert in the window was announced inside quiet hours, so no hold was exercised. The hold path exists in `compute_pending_deliveries`; first overnight announcement will produce direct evidence (would_deliver `created_at` local ≥ 06:00).
4	Rate cap ≤ 3/h per user	[OK]	max would-deliver events per user-hour = **3** (= cap), across 145 would-deliver rows / 53 users.
5	Stability — tick every 10 min, errors rare/non-fatal	[OK]	132 consecutive ticks, gap median **10.0 min, max 10.0 min**, zero gaps; poll payloads report **0 errors**; per-source failure isolation verified earlier (doc 04 §4).
6	No cross-poll duplicates	[OK]	829 duplicate detections swallowed across 132 cycles — re-polls stored nothing new (idempotent `source_ref` + 48 h title-Jaccard).

Overall: 5/6 criteria pass, 1 untested-by-circumstance (no overnight announcement yet). No criterion failed.

3. Detection volume in window

• 20 stored alerts (new since 10-03): 16 JBALB + 4 Telegram; 14 unscheduled / 6 scheduled; **all water**; locations: Betong ×4, Samarahan/Asajaya ×2, Serian ×2, Sarikei ×2, Sibu/Selangau, Sri Aman, Kapit/Song, Mukah/Dalat, Bintulu/Sebauh + 2 Telegram Sibu notices.

- 145 would-deliver computations (53 opted-in users × matched alerts, first-due only).

4. Stored alerts for the precision spot-check (owner review)

Dates are announcement time (Asia/Kuching). All titles are official interruption notices; no off-topic or non-Sarawak content in the sample.

Source	Kind	Announced	Title (truncated)
jbalb	scheduled	10-16 08:00	Scheduled Water Interruption Notice at Serian (Gedong)
telegram	scheduled	10-12 16:08	Scheduled Water Supply Interruption Notice
telegram	unscheduled	10-12 14:59	Unscheduled Water Supply Interruption
jbalb	scheduled	10-12 08:00	Scheduled Water Interruption Notice at Samarahan (Asajaya)
jbalb	unscheduled	10-10 08:00	Unscheduled Water Interruption Notice at Betong (Spaoh, Debak)
jbalb	unscheduled	10-10 08:00	Unscheduled Water Interruption Notice at Samarahan (Sebuyau)
jbalb	unscheduled	10-10 08:00	Unscheduled Water Interruption Notice at Kapit (Song)
jbalb	scheduled	10-09 08:00	Scheduled Water Interruption Notice at Serian
jbalb	unscheduled	10-09 08:00	Unscheduled Water Interruption Notice at Sri Aman
jbalb	unscheduled	10-09 08:00	Unscheduled Water Interruption Notice at Betong (Saratok, Pusa, Debak)
jbalb	unscheduled	10-09 08:00	Unscheduled Water Interruption Notice at Mukah (Dalat)
jbalb	scheduled	10-08 08:00	Scheduled Water Interruption Notice at Sarikei
telegram	unscheduled	10-06 15:07	Unscheduled Water Supply Interruption
telegram	unscheduled	10-06 09:20	Interconnection of 12" AC Pipe at Jalan Desa…
jbalb	unscheduled	10-05 08:00	Unscheduled Water Interruption Notice at Sibu (Selangau)
jbalb	unscheduled	10-05 08:00	Unscheduled Water Interruption Notice at Betong (Spaoh, Debak)
jbalb	scheduled	10-05 08:00	Scheduled Water Interruption Notice at Bintulu (Sebauh)
jbalb	unscheduled	10-05 08:00	Unscheduled Water Interruption Notice at Betong (Kabong, Roban)
jbalb	unscheduled	10-04 08:00	Unscheduled Water Interruption Notice at Sarikei (Meradong)
jbalb	unscheduled	10-03 08:00	Unscheduled Water Interruption Notice at Samarahan (Asajaya)

5. Stage 2 checklist status (doc 04 §6)

#	Item	Status
1	Shadow evidence review	[OK] this document (owner to confirm)
2	`/settings` toggle for `wants_urgent_alerts`	■ **recommended before go-live** — without it, 53 users are opted-in by default with no self-service off switch
3	Incident clustering for news watch	■ open (prevents duplicate alerts when multiple outlets cover one burst)
4	Restoration threading	■ open (link "restored" follow-ups to the original notice)
5	rss.app FB bridge (optional, paid)	■ owner decision deferred
6	`UTILITY_ALERT_SHADOW_MODE=false` + redeploy	■ **owner decision, after items 2–4**

6. Recommendation

Evidence supports **go** on detection quality: precision is perfect on the sample, delivery math (rate cap, per-user skip, zero pushes) is verified, and the pipeline is stable at full cadence. However, **do not flip** `SHADOW_MODE=false` **yet**: ship the `/settings` opt-out toggle first (item 2) so users can control alerts before any push reaches them, and ideally item 3 (clustering) so the first live burst doesn't arrive as 3–4 duplicate alerts from different outlets. Items 3–4 can ship after go-live as refinements if the owner prefers an early live date limited to Telegram+JBALB sources.

Suggested order: settings toggle → flip shadow off (Telegram+JBALB only) → clustering + restoration threading in Week 3.